

Clinical Case Bank

43 exam-style cases built from the *Integrated Clinical Revision Guide* (Ain Shams 2025–2026, Levels A–C)

How to use: each case is worth **10 marks**. Answer the case page on its own, then turn over: the next page has the model answer and the marking scheme. Doses are as printed in the source guide, so check them against your department's textbook.

Part 1 · Cardiology (6 cases)

- 1 Crushing chest pain with hypotension (inferior STEMI + RV infarct)
- 2 Sudden breathlessness after a long flight (pulmonary embolism)
- 3 Fever and a louder murmur after a dental extraction (infective endocarditis)
- 4 Breathless at night with pink frothy sputum (acute heart failure)
- 5 Irregular palpitations in an elderly woman (atrial fibrillation)
- 6 Child with flitting joint pains and a new murmur (rheumatic fever)

Part 2 · Chest (6 cases)

- 7 Breathless woman who can only say single words (acute severe asthma)
- 8 Smoker with worse breathlessness and green sputum (COPD exacerbation)
- 9 Elderly man with fever, confusion and consolidation (pneumonia)
- 10 Tall young man with sudden chest pain (pneumothorax)
- 11 Progressive breathlessness with Velcro crackles (pulmonary fibrosis)
- 12 Chronic cough, night sweats and an apical cavity (pulmonary TB)

Part 3 · Hematology & Clinical Pathology (6 cases)

- 13 Tired woman with heavy periods (iron-deficiency anemia)
- 14 Macrocytic anemia with numb feet (B12 deficiency)
- 15 Petechiae in a well young woman (ITP)
- 16 Child with severe bone pain and known HbSS (sickle cell crisis)
- 17 Boy with a hot swollen knee after minor trauma (hemophilia A)
- 18 Fever and back pain minutes into a blood transfusion (transfusion reaction)

Part 4 · Tropical Medicine & Infectious Diseases (6 cases)

- 19 Fever, neck stiffness and a rash (meningococcal meningitis)
- 20 Undulant fever in a farmer (brucellosis)
- 21 Right upper quadrant pain after bloody diarrhea (amoebic liver abscess)
- 22 Jaundiced man with abnormal hepatitis serology (hepatitis B)
- 23 Canal swimmer with terminal hematuria (schistosomiasis)
- 24 Student with sore throat and a rash after ampicillin (infectious mononucleosis)

Part 5 · Immunology & Allergy (5 cases)

- 25 Collapse after an antibiotic injection (anaphylaxis)
- 26 Swollen lips and abdominal colic with no hives (hereditary angioedema)
- 27 Daily hives for 3 months (chronic spontaneous urticaria)
- 28 Boy with repeated chest infections since infancy (primary immunodeficiency)

29 Blisters and sore lips after a new drug (SJS/TEN)

Part 6 · Dermatology, Venereology & Andrology (8 cases)

30 Painless penile ulcer (primary syphilis)

31 Itchy family at night (scabies)

32 Profuse urethral pus 3 days after exposure (gonococcal urethritis)

33 Child with a scaly bald patch on the scalp (tinea capitis)

34 Child with honey-coloured crusts around the nose (impetigo)

35 Teenager with nodulocystic acne and scarring (acne vulgaris)

36 Painful band of blisters on one side of the chest (herpes zoster)

37 Couple unable to conceive for 2 years (male infertility)

Part 7 · Psychiatry (6 cases)

38 "They'd be better off without me" (depression + suicide risk)

39 Young man hearing voices for 8 months (schizophrenia)

40 No sleep for a week and spending all his savings (bipolar mania + lithium)

41 Sudden attacks of chest pain and fear of dying (panic disorder)

42 Very thin teenager who thinks she is fat (anorexia nervosa)

43 Confused and shaking on day 3 after surgery (delirium tremens)

Case 1

Cardiology · 10 marks

A 58-year-old male smoker has had 2 hours of crushing retrosternal pain with sweating. BP 85/50 mmHg, pulse 54/min, JVP raised, lungs clear. ECG: ST elevation in II, III and aVF. The nearest primary-PCI centre is 3 hours away.

1. What is the diagnosis? Name the likely culprit artery and the extra ECG lead you would request (and why). **[2]**
2. Mention two investigations apart from the ECG and what each adds. **[1]**
3. Write the immediate drug treatment with doses. **[3]**
4. Which reperfusion strategy is appropriate here, and why? List 3 **absolute** contraindications to it. **[2]**
5. **Special:** Which drugs must be avoided in this patient, and how would you treat his hypotension? **[2]**

Case 1 — Answers & marking scheme

Q	Model answer	Marks
1	Inferior STEMI complicated by right-ventricular infarction (hypotension, ↑JVP, clear lungs). Culprit: RCA (~85%; otherwise LCx). Request right-sided leads (V4R) : ST↑ in V4R confirms RV infarct (proximal RCA).	1+0.5+0.5
2	Any two: serial troponin I/T (rises 3–6 h, peak 36–48 h); echo (wall motion, RV dilatation, mechanical complications); CBC, kidney function, electrolytes, lipids. Do not delay reperfusion waiting for results.	0.5 each
3	• Aspirin 162–325 mg chewed, then 75–100 mg daily • Clopidogrel 300–600 mg load → 75 mg daily (or Ticagrelor 180 mg → 90 mg bid) • Anticoagulant: Enoxaparin 30 mg IV bolus then 1 mg/kg SC q12h (<75 y), or UFH 60 U/kg IV (max 4000) → 12 U/kg/h • Atorvastatin 80 mg; O₂ only if hypoxic; Morphine 2–4 mg IV for refractory pain • Bradycardia: Atropine 0.5–1 mg IV (max 3 mg). Withhold β-blocker (shock, bradycardia).	0.5 each (max 3)
4	Fibrinolysis (PCI is >120 min away): door-to-needle ≤30 min, e.g. Tenecteplase (weight-based) or Alteplase 15 mg → 50 mg/30 min → 35 mg/60 min, or Streptokinase 1.5 MIU/60 min; then transfer for angiography. Absolute CI (any 3): prior intracranial hemorrhage · cerebral AVM · intracranial malignancy · ischemic stroke <3 months · suspected aortic dissection · active bleeding/bleeding diathesis · significant head/face trauma <3 months.	1 + 1
5	RV infarct is preload-dependent: avoid nitrates and diuretics (also PDE-5 inhibitors with nitrates). Treat hypotension with IV normal-saline boluses ; inotropes if no response.	1 + 1

Pearl: Inferior STEMI + hypotension + clear lungs = RV infarct until proven otherwise. Giving nitroglycerin here can crash the BP.

Case 2

Cardiology · 10 marks

A 34-year-old woman on the combined oral contraceptive pill develops sudden pleuritic chest pain and breathlessness 3 days after a 12-hour flight. Her left calf is swollen and tender. Pulse 112/min, BP 118/74 mmHg, RR 26/min, SpO₂ 91% on air. She has no previous VTE, no cancer and no hemoptysis.

1. Calculate her Wells score and interpret it (two-tier). **[2]**
2. What is the next diagnostic test? When would a D-dimer be the right first test instead? **[2]**
3. CTPA confirms PE. Echo shows RV/LV ratio 1.2 and troponin is raised. Classify the PE and give the treatment with doses. **[3]**
4. Two hours later her BP falls to 82/50 mmHg. What do you do? **[1]**
5. **Special:** How long should she be anticoagulated? Name one situation where warfarin is mandatory and one where LMWH is preferred. **[2]**

Case 2 — Answers & marking scheme

Q	Model answer	Marks
1	Clinical signs of DVT 3 + PE most likely diagnosis 3 + HR >100 1.5 = 7.5 → PE likely (>4) . (Immobilisation counts only if ≥3 days in bed or surgery <4 wk.)	1 + 1
2	CTPA directly (PE likely). D-dimer first only when PE is unlikely (Wells ≤4) : a negative D-dimer (<500 ng/mL) excludes PE; a positive one → CTPA.	1 + 1
3	Submassive (intermediate-risk) PE : normotensive with RV strain and ↑troponin. Enoxaparin 1 mg/kg SC q12h (or 1.5 mg/kg daily) or Fondaparinux 7.5 mg ; monitor closely for deterioration. (Low-risk PE would get a DOAC: Apixaban 10 mg bid ×7 d → 5 mg bid, or Rivaroxaban 15 mg bid ×21 d → 20 mg daily.)	1 + 2
4	Now massive (high-risk) PE → rescue thrombolysis : Alteplase 100 mg over 2 h with UFH 80 U/kg + 18 U/kg/h. If contraindicated or failed → catheter/surgical embolectomy.	1
5	Provoked (flight + OCP): at least 3 months ; stop the combined OCP. Unprovoked or persistent risk → extended/indefinite. Warfarin (INR 2–3) mandatory in antiphospholipid syndrome ; LMWH preferred in cancer and pregnancy .	1 + 1

Pearl: IVC filter only if anticoagulation is contraindicated or PE recurs despite treatment.

Case 3

Cardiology · 10 marks

A 45-year-old man with known rheumatic mitral regurgitation had a dental extraction 4 weeks ago without antibiotic cover. He now has 3 weeks of fever 38.6 °C, weight loss, splinter hemorrhages, painful nodules on his finger pads and a louder MR murmur. Urine shows microscopic hematuria.

1. What is the likely diagnosis and organism? Name the painful finger nodules and one other immunologic sign. **[2]**
2. How should blood cultures be taken? Which echo comes first, and when is TEE needed? **[2]**
3. State the modified Duke criteria needed for a **definite** diagnosis. **[2]**
4. Give the empirical antibiotic regimen for native-valve IE with doses and duration. **[2]**
5. **Special:** List 3 indications for urgent surgery. Who should receive dental prophylaxis, and with what? **[2]**

Case 3 — Answers & marking scheme

Q	Model answer	Marks
1	Subacute infective endocarditis on a rheumatic mitral valve, most likely viridans streptococci (dental source). Painful finger-pad nodules = Osler nodes . Other immunologic signs: Roth spots, glomerulonephritis (hematuria), positive RF.	1 + 0.5 + 0.5
2	≥3 sets from separate sites , ≥30–60 min between first and last, before antibiotics. TTE first ; TEE if high risk, prosthetic valve or suspected abscess (or TTE negative with high suspicion).	1 + 1
3	Definite: 2 major , or 1 major + 3 minor , or 5 minor . Major = typical organism in 2 separate cultures / persistent bacteremia; echo vegetation, abscess or new regurgitation. (Possible: 1 major + 1 minor, or 3 minor.)	1 + 1
4	IV for 4–6 weeks: Ampicillin-sulbactam 12 g/day in 4 doses + Gentamicin 3 mg/kg/day , or Vancomycin 15–20 mg/kg q8–12h + Gentamicin . Adjust to culture sensitivity.	1.5 + 0.5
5	Surgery (any 3): heart failure from acute severe regurgitation · uncontrolled infection (bacteremia >7–10 d, abscess, fistula, new heart block, fungal, enlarging vegetation) · embolism prevention (vegetation >10 mm after an embolus). Prophylaxis only for high-risk patients (prosthetic valve, prior IE, unrepaired cyanotic CHD, repaired CHD with residual defect): Amoxicillin 2 g PO 30–60 min before ; allergy: Clindamycin 600 mg.	1 + 1

Pearl: Early prosthetic-valve IE (<12 months): Vancomycin + Gentamicin + Rifampin 300 mg q8h.

Case 4

Cardiology · 10 marks

A 66-year-old man with a previous anterior MI wakes at 3 a.m. gasping, sitting up, coughing pink frothy sputum. He is sweaty and warm. BP 170/100 mmHg, pulse 118/min, SpO₂ 86% on air, bilateral crackles to the mid-zones, S3 gallop, raised JVP, pitting ankle edema.

1. What is the diagnosis? Classify his hemodynamic profile (Nohria-Stevenson). **[2]**
2. List 3 investigations and what each shows. **[1.5]**
3. Write the immediate management in order, with doses. **[3]**
4. When would you stop or avoid nitrates? **[1]**
5. **Special:** Echo later shows EF 30%. Name the 4 pillars of long-term therapy (one drug + dose each) and one precaution when switching from an ACE inhibitor to an ARNI. **[2.5]**

Case 4 — Answers & marking scheme

Q	Model answer	Marks
1	Acute cardiogenic (flash) pulmonary edema / acute decompensated HF. Profile: warm & wet (the commonest).	1 + 1
2	Any 3: BNP/NT-proBNP (NT-proBNP <125 rules out HF) · CXR (bat-wing edema, Kerley B, cardiomegaly, effusions) · ECG / troponin (ischemia, arrhythmia) · echo (EF, valves) · ABG, kidney function, electrolytes.	0.5 each
3	Sit upright, legs dangling O₂ if SpO₂ <90% → CPAP/BiPAP (PEEP 5–10 cm H₂O) Furosemide 40–80 mg IV (check urine output >100–150 mL/h at 2 h) - SBP >110: IV nitroglycerin 10–20 → 100–200 mcg/min	0.75 each
4	Avoid/hold nitrates if SBP <90, RV infarct or PDE-5 inhibitor use. SBP <90 = cardiogenic shock (cold & wet): dobutamine 2–20 mcg/kg/min or milrinone ± norepinephrine.	1
5	Quadruple therapy: ARNI sacubitril/valsartan 24/26–49/51 mg bid → 97/103 mg bid · β-blocker bisoprolol 1.25 → 10 mg daily · MRA spironolactone 12.5–25 → 50 mg (hold if K ⁺ >5.0 or eGFR <30) · SGLT2i dapagliflozin/empagliflozin 10 mg daily. Plus loop diuretic to dry weight. ACEI → ARNI: 36-hour washout (angioedema risk).	0.5 each

Case 5

Cardiology · 10 marks

A 76-year-old woman with hypertension and type 2 diabetes has had palpitations and mild breathlessness for "about a week". Pulse 138/min, irregularly irregular; BP 132/80 mmHg; she is alert with no chest pain. ECG: narrow QRS, no P waves, irregular RR intervals.

1. What is the diagnosis? List 3 investigations to look for a cause. **[2]**
2. Is she stable or unstable? Give the acute rate-control drug with dose. **[2]**
3. Can she be cardioverted today? Explain. **[2]**
4. Calculate her CHA₂DS₂-VASc score and state the anticoagulation decision. **[2]**
5. **Special:** What would you do if she became hypotensive and confused? Which drug is dangerous in pre-excited AF (WPW)? **[2]**

Case 5 — Answers & marking scheme

Q	Model answer	Marks
1	Atrial fibrillation with rapid ventricular response. Any 3: TSH (thyrotoxicosis), electrolytes, echo (valve disease, LA size, LV function), troponin, CBC, glucose, kidney function.	1 + 1
2	Stable (no hypotension, ischemic pain, ↓ consciousness or pulmonary edema). Rate control: Metoprolol 2.5-5 mg IV or Diltiazem 0.25 mg/kg IV (long-term: β-blocker, non-DHP CCB or digoxin).	1 + 1
3	No. AF for >48 h or unknown duration → ≥3 weeks of anticoagulation or a TEE to exclude LA thrombus before cardioversion (risk of stroke). (If <48 h: rhythm control, e.g. amiodarone 300 mg over 1 h.)	2
4	HTN 1 + DM 1 + age ≥75 = 2 + female 1 = 5 . Anticoagulate (≥3 in women, ≥2 in men): DOAC preferred .	1 + 1
5	Unstable → synchronised DC cardioversion immediately. Adenosine (and AV-nodal blockers such as verapamil/digoxin) is dangerous in pre-excited AF.	1 + 1

Pearl: Rate control suits elderly or asymptomatic patients; rhythm control (amiodarone, flecainide, ablation) suits symptomatic, young or HF patients.

Case 6

Cardiology · 10 marks

A 12-year-old girl (35 kg) from rural Upper Egypt had a sore throat 3 weeks ago. She now has fever 38.4 °C and painful swollen knees, then elbows, each lasting a few days. There is a new apical pansystolic murmur radiating to the axilla. ESR 75 mm/h, CRP 5 mg/dL, ECG PR interval 220 ms.

1. What is the diagnosis? List the criteria she fulfils. **[3]**
2. Which investigations prove a recent streptococcal infection, and which assess the heart? **[2]**
3. Outline her acute treatment with doses (eradication + anti-inflammatory). Echo shows moderate MR. **[3]**
4. **Special:** Give the secondary prophylaxis regimen and how long it should continue. **[2]**

Case 6 — Answers & marking scheme

Q	Model answer	Marks
1	Acute rheumatic fever (first attack) with carditis. Jones criteria (moderate/high-risk population): Major: carditis (new MR murmur) + migratory polyarthrititis . Minor: fever $\geq 38^{\circ}\text{C}$, ESR ≥ 30 / CRP ≥ 3 (a long PR does not count as minor when carditis is already a major criterion). Plus evidence of preceding GAS infection → 2 major criteria = diagnostic.	1 + 1 + 1
2	GAS evidence: throat culture, rapid antigen test, rising ASO titre (anti-DNase B). Heart: echocardiography (valve lesion, effusion, LV function), ECG, CXR (cardiomegaly).	1 + 1
3	- Eradicate GAS: Benzathine penicillin G 1.2 MU IM once (she is >27 kg) or Penicillin V 500 mg bid ×10 d; allergy: Erythromycin 250 mg qid ×10 d. - Moderate–severe carditis: Prednisone 1–2 mg/kg/day (max 60 mg) for 2–4 weeks, then taper while adding aspirin. (Arthritis only would be Aspirin 80–100 mg/kg/day in 4 doses for 4–6 weeks.) Bed rest, treat heart failure if present.	1 + 1.5 + 0.5
4	Benzathine penicillin 1.2 MU IM every 3–4 weeks (or Penicillin V 250 mg bid; Sulfadiazine 1 g daily; allergy Erythromycin 250 mg bid). Duration: carditis with persistent valve disease → 10 years or until age 40 (whichever longer, often lifelong). If MR resolves → 10 years or until 21.	1 + 1

Pearl: Prophylaxis durations: no carditis 5 y / until 21 · carditis, no residual disease 10 y / until 21 · residual valve disease 10 y / until 40.

Case 7

Chest · 10 marks

A 24-year-old woman with asthma, who uses only a salbutamol inhaler, arrives breathless. She can speak only single words, is using accessory muscles, RR 34/min, pulse 128/min, SpO₂ 88% on air, PEF 40% of predicted, loud wheeze.

1. Classify the severity of this attack, giving 2 features that support it. **[2]**
2. Write the immediate management in the ED with doses. **[3]**
3. One hour later she is drowsy with a quiet chest. ABG: PaCO₂ 46 mmHg. Interpret this and state what you do next. **[2]**
4. How is asthma confirmed on spirometry when she is stable? **[1]**
5. **Special:** What is wrong with her current maintenance treatment, and what is the preferred GINA approach? **[2]**

Case 7 — Answers & marking scheme

Q	Model answer	Marks
1	Acute severe asthma. Any 2: single words · RR >30 · HR >120 · SpO ₂ <90% · PEF ≤50% · accessory muscles.	1 + 1
2	• O₂ to SpO₂ 93-95% • Salbutamol 2.5-5 mg nebulised (or 4-10 puffs via spacer) every 20 min for 1 h • Ipratropium 0.5 mg nebulised with salbutamol (×3 doses) • Prednisolone 40-50 mg PO ×5-7 d or Hydrocortisone 100 mg IV q6h • No response at 1 h → MgSO₄ 2 g IV over 20 min (aminophylline 5 mg/kg load reserved)	0.5 each + 1 for MgSO₄
3	Life-threatening asthma: a "normal"/raised PaCO ₂ (>42 mmHg) in a breathless asthmatic means she is tiring and arrest is near (silent chest, drowsiness). → ICU, prepare for intubation (↓consciousness, exhaustion, pH <7.30, PaCO ₂ >45).	1 + 1
4	FEV ₁ /FVC reduced (<0.75-0.80) with bronchodilator reversibility: FEV₁ ↑ >12% and >200 mL after 200-400 mcg salbutamol (or PEF variability >10%).	1
5	SABA alone (without ICS) is contraindicated. Preferred (Track 1): ICS-formoterol as reliever ± maintenance, stepped up by severity (after this attack at least step 3-4: low/medium-dose ICS-formoterol maintenance + reliever). Check inhaler technique and triggers.	1 + 1

Case 8

Chest · 10 marks

A 68-year-old man (50 pack-years) with COPD has had 3 days of increased breathlessness and more, greener sputum. RR 28/min, SpO₂ 82% on air, widespread wheeze. In the ambulance he was given 15 L/min O₂ via a non-rebreather mask and is now drowsy. ABG: pH 7.29, PaCO₂ 68 mmHg, PaO₂ 95 mmHg.

1. Interpret the ABG and explain why he became drowsy. **[2]**
2. How should oxygen be given in this patient (device, %, target)? **[1]**
3. Write the drug treatment of the exacerbation with doses and durations. **[3]**
4. When is NIV indicated, and when does he need intubation? **[2]**
5. **Special:** What diagnoses COPD on spirometry? Name the single most effective measure to slow FEV₁ decline and 2 vaccines. **[2]**

Case 8 — Answers & marking scheme

Q	Model answer	Marks
1	Acute-on-chronic type II (hypercapnic) respiratory failure with respiratory acidosis. Uncontrolled high-flow O ₂ → CO ₂ retention (loss of hypoxic drive, V/Q worsening, Haldane effect) → CO₂ narcosis .	1 + 1
2	Controlled O₂ via 24-28% Venturi mask, target SpO₂ 88-92%.	1
3	• Salbutamol 2.5-5 mg + Ipratropium 0.5 mg nebulised q4-6h • Prednisone 40 mg daily for exactly 5 days • Antibiotics ×5 days (≥2 Anthonisen criteria incl. purulence): Amox-clav 875/125 bid, or Azithromycin 500 mg, or Doxycycline 100 mg bid	1 each
4	NIV (BiPAP) first-line if pH ≤7.35 and PaCO ₂ ≥45 despite therapy. Intubate if GCS <8, exhaustion, pH <7.25 or NIV failure.	1 + 1
5	Post-bronchodilator FEV₁/FVC <0.70. Stopping smoking is the #1 measure. Vaccines: influenza + pneumococcal (PCV20/PPSV23). (Also pulmonary rehab, LTOT in very severe disease.)	0.5 + 0.5 + 1

Case 9

Chest · 10 marks

A 72-year-old man has had 2 days of fever, productive cough with rusty sputum and right-sided pleuritic pain. Today he is newly confused. RR 32/min, BP 95/55 mmHg, pulse 115/min, temperature 39.2 °C, bronchial breathing at the right base. Urea 9 mmol/L. CXR: right lower-lobe consolidation.

1. What is the diagnosis and most likely organism? **[1]**
2. Calculate his CURB-65 score, interpret it, and say where he should be managed. **[3]**
3. List 3 investigations apart from the CXR. **[1.5]**
4. Give the empirical antibiotics for his severity with doses. What would you give a healthy 30-year-old outpatient? **[2.5]**
5. **Special:** If he had developed this pneumonia on day 5 of a hospital admission, what would change? **[2]**

Case 9 — Answers & marking scheme

Q	Model answer	Marks
1	Severe community-acquired pneumonia , likely <i>Streptococcus pneumoniae</i> .	1
2	Confusion 1 + Urea >7 1 + RR ≥30 1 + BP (diastolic ≤60) 1 + age ≥65 1 = 5. Mortality 14–28% → hospital, consider ICU (4–5).	1 + 1 + 1
3	Any 3: blood cultures before antibiotics · sputum Gram stain and culture · CBC, CRP · kidney function/electrolytes · ABG · urinary pneumococcal/Legionella antigen.	0.5 each
4	Severe (ICU): Ceftriaxone 2 g + Azithromycin 500 mg IV (Pseudomonas risk: Pip-tazo 4.5 g q6h or Cefepime 2 g q8h; MRSA risk: add Vancomycin or Linezolid). (Ward, CURB 2–3: Ceftriaxone 1–2 g + Azithromycin, or Levofloxacin 750 mg.) Healthy outpatient (0–1): Amoxicillin 1 g tid or Doxycycline 100 mg bid or Azithromycin, for ≥5 days (afebrile 48 h).	1.5 + 1
5	Hospital-acquired pneumonia (≥48 h after admission): cover resistant Gram-negatives and MRSA → Pip-tazo 4.5 g q6h or Meropenem 1 g q8h + Vancomycin ; de-escalate at 48–72 h on culture results.	2

Case 10

Chest · 10 marks

A tall, thin 22-year-old smoker develops sudden left-sided chest pain and breathlessness at rest. He is otherwise well. RR 22/min, SpO₂ 95%, reduced breath sounds and hyper-resonance on the left. CXR: left pneumothorax with a 2.5 cm rim at the hilum.

1. What is the diagnosis and its usual cause? **[1]**
2. How do you manage him? How would you manage a rim <2 cm with no symptoms? **[2]**
3. How would management differ in a 65-year-old with COPD? **[1]**
4. **Special:** A trauma patient has severe distress, hypotension, distended neck veins and tracheal deviation away from the silent side. What is it, and what do you do immediately (exact site)? **[3]**
5. Give 3 indications for referral to thoracic surgery in pleural disease. **[3]**

Case 10 — Answers & marking scheme

Q	Model answer	Marks
1	Primary spontaneous pneumothorax: rupture of apical blebs (tall thin young males, smoking).	1
2	≥ 2 cm: needle aspiration , then chest drain if aspiration fails . < 2 cm and asymptomatic: observe (repeat CXR, safety advice).	1 + 1
3	Secondary pneumothorax (COPD, TB): chest tube is mandatory .	1
4	Tension pneumothorax: clinical diagnosis, do not wait for a CXR. Immediate needle decompression in the 2nd intercostal space, mid-clavicular line (or 5th ICS anterior axillary line) → chest tube in the 5th ICS mid-axillary line with underwater seal .	1 + 1 + 1
5	Any 3: persistent air leak > 3-5 days · second (recurrent) pneumothorax · multiloculated empyema failing drainage at 48-72 h · thick peel/trapped lung · hemothorax > 1500 mL or > 200 mL/h .	1 each

Case 11

Chest · 10 marks

A 67-year-old retired man has had 18 months of slowly worsening exertional breathlessness and a dry cough. He has finger clubbing and fine bibasal end-inspiratory "Velcro" crackles. SpO₂ 95% at rest, falling to 86% on walking.

1. What is the most likely diagnosis? **[1]**
2. What pattern do you expect on pulmonary function tests (4 findings)? **[2]**
3. Describe the HRCT findings of a UIP pattern. **[2]**
4. Name the 2 antifibrotic drugs with doses, and 3 supportive measures. **[3]**
5. **Special:** How do you tell this patient's breathlessness from COPD and from heart failure at the bedside? **[2]**

Case 11 — Answers & marking scheme

Q	Model answer	Marks
1	Idiopathic pulmonary fibrosis (interstitial pulmonary fibrosis).	1
2	Restrictive: ↓TLC, ↓FVC, FEV₁/FVC normal or >0.70 , markedly ↓ DLCO ; exertional desaturation.	0.5 each
3	Subpleural, basal reticulation, honeycombing, traction bronchiectasis , little ground-glass.	2
4	Pirfenidone titrated to 801 mg tid with meals · Nintedanib 150 mg bid with food (both slow FVC decline). Supportive: O₂ , pulmonary rehabilitation , early lung-transplant referral (also vaccines, stop smoking).	1 + 1 + 1
5	IPF: Velcro crackles + clubbing , restrictive. COPD: smoker with morning cough + wheeze , obstructive spirometry. HF: PND, orthopnea, S3 , raised JVP, edema.	1 + 1

Case 12

Chest · 10 marks

A 35-year-old man (60 kg) has had a cough for 5 weeks with streaky hemoptysis, evening fever, night sweats and 6 kg weight loss. CXR: right upper-lobe cavity.

1. List the investigations to confirm the diagnosis, saying what each offers. **[3]**
2. Write his treatment regimen with doses for his weight. **[3]**
3. Match the side effect to the drug: (a) orange urine, (b) red-green colour blindness, (c) acute gout, (d) burning feet. Which supplement prevents (d)? **[2]**
4. **Special:** Define treatment failure and MDR-TB. **[2]**

Case 12 — Answers & marking scheme

Q	Model answer	Marks
1	• Sputum ZN smear ×3 for acid-fast bacilli • GeneXpert MTB/RIF: result in 2 h and detects rifampicin resistance • Culture (LJ or MGIT) = gold standard; gives drug sensitivity • Also: HIV test, baseline LFT, visual acuity / colour vision (before ethambutol)	0.75 each
2	2 months HRZE then 4 months HR under DOTS (6 months total). For 60 kg: Isoniazid 300 mg (5 mg/kg, max 300) · Rifampicin 600 mg (10 mg/kg) · Pyrazinamide 1.5 g (25 mg/kg) · Ethambutol 900 mg (15 mg/kg) + Pyridoxine.	1 + 2
3	(a) Rifampicin · (b) Ethambutol (optic neuritis) · (c) Pyrazinamide (hyperuricemia) · (d) Isoniazid (neuropathy) → Pyridoxine 10-50 mg/day .	0.4 each
4	Treatment failure : smear still positive at ≥5 months of treatment. MDR-TB : resistant to at least isoniazid + rifampicin (needs 9-20 months of second-line therapy).	1 + 1

Pearl: Rifampicin is a CYP450 inducer, so warn about OCP failure and interactions with warfarin and ART. Streptomycin is contraindicated in pregnancy.

Case 13

Hematology · 10 marks

A 32-year-old woman (60 kg) with heavy periods complains of fatigue and craves ice. She has spoon-shaped nails and angular cheilitis. Hb 8 g/dL, MCV 68 fL, RBC $3.9 \times 10^{12}/L$, RDW high; the smear shows pencil cells.

1. Give 2 differential diagnoses of microcytic anemia and show how iron studies separate them. **[3]**
2. Calculate the Mentzer index and interpret it. **[1]**
3. Write her oral iron treatment (drug, dose, duration, how to take it) and how you check that she is responding. **[3]**
4. **Special:** Give 2 indications for parenteral iron and calculate her total iron deficit (Ganzoni), target Hb 15 g/dL. **[3]**

Case 13 — Answers & marking scheme

Q	Model answer				Marks
1		Iron deficiency	Chronic disease	β-thal trait	1.5 each
	Ferritin	↓↓ (<15–30)	N / ↑	N–↑	
	TIBC	↑ (>360)	↓ (<240)	N	
	TSAT	<16%	15–20% / N	N	
	Other	RDW ↑	marrow iron present	HbA2 >3.5%, target cells	
	Any two conditions correctly contrasted.				
2	MCV ÷ RBC = $68 \div 3.9 \approx 17.4$ (>13) → favours iron deficiency (<13 suggests thalassemia trait).				1
3	Ferrous sulfate 200–325 mg three times daily (65 mg elemental iron/tab), on an empty stomach 1 h before meals with vitamin C. Continue 3–6 months after Hb normalises to refill stores. Response: reticulocytes peak on days 7–10 , Hb rises 1–2 g/dL every 2–3 weeks. Also treat the cause (menorrhagia → gynaecology).				1 + 1 + 1
4	Indications (any 2): intolerance · poor compliance · malabsorption (e.g. celiac) · ongoing bleeding exceeding oral replacement. Ganzoni = wt × (target – actual Hb) × 2.4 + 500 = 60 × (15 – 8) × 2.4 + 500 = 1008 + 500 = ≈1500 mg .				1 + 2

Pearl: Ferritin is an acute-phase reactant. A level of 30–100 can still mean iron deficiency when there is inflammation.

Case 14

Hematology · 10 marks

A 55-year-old woman with hypothyroidism has fatigue, a lemon-yellow tint and tingling in both feet. Examination: loss of vibration sense and joint-position sense in the legs, positive Romberg test. Hb 8 g/dL, MCV 115 fL, WBC and platelets mildly low, reticulocytes low, LDH raised.

1. What is the most likely diagnosis and cause? Name the neurological complication. **[2]**
2. List the investigations that confirm it, including one that distinguishes it from folate deficiency. **[3]**
3. Write the treatment with doses for this patient. **[3]**
4. **Special:** A junior doctor wants to start folic acid while the results are pending. Why is this dangerous? **[2]**

Case 14 — Answers & marking scheme

Q	Model answer	Marks
1	Vitamin B12 deficiency megaloblastic anemia due to pernicious anemia (autoimmune; linked to thyroid disease). Neuro: subacute combined degeneration of the cord .	1 + 1
2	• Smear: macro-ovalocytes, hypersegmented neutrophils (≥ 5 lobes), pancytopenia, low retics • Serum B12 ↓; ↑ methylmalonic acid (MMA) + ↑ homocysteine (folate deficiency: MMA normal) • Anti-intrinsic factor / anti-parietal cell antibodies • Marrow (if done): hypercellular, megaloblastic	0.75 each
3	With neurological involvement: Hydroxocobalamin/B12 1000 µg IM on alternate days for 3 weeks , then every 3 months for life . (Without neuro: 1000 µg IM daily ×7–14 d → weekly ×4–8 wk → every 3 months.) Watch for hypokalemia early in treatment. Retic response on days 3–7.	2 + 1
4	Folic acid alone corrects the blood count but lets the neurological damage (SCD) progress irreversibly . Always check and replace B12 first.	2

Case 15

Hematology · 10 marks

A 24-year-old woman presents with petechiae on her legs, gum bleeding and epistaxis for 3 days. She is otherwise well and has no splenomegaly or lymphadenopathy. Platelets $8 \times 10^9/L$ (large platelets on the smear); Hb, WBC, PT and aPTT normal.

1. What is the most likely diagnosis? Give 3 conditions you must exclude. **[2]**
2. What do the bleeding time, PT and aPTT show, and how does this differ from hemophilia and DIC? **[2]**
3. Write her emergency management with doses. **[3]**
4. When would you transfuse platelets in this condition? **[1]**
5. **Special:** She relapses on tapering steroids. List the second-line options. **[2]**

Case 15 — Answers & marking scheme

Q	Model answer	Marks
1	Immune thrombocytopenia (ITP): IgG anti-GPIIb/IIIa. Exclude (any 3): pseudothrombocytopenia (EDTA clumping → repeat in citrate) · TTP (schistocytes) · leukemia/aplastic anemia · drugs · HIV/HCV · SLE.	1 + 1
2	ITP: platelets ↓, BT ↑, PT and aPTT normal . Hemophilia: normal platelets and BT, isolated ↑aPTT . DIC: everything abnormal (↓plt, ↑PT, ↑aPTT, ↑TT, ↓fibrinogen, ↑D-dimer, schistocytes).	1 + 1
3	Platelets <10 ×10 ⁹ /L with bleeding = emergency: IV methylprednisolone + IVIG 1 g/kg/day ×2 days (or 0.4 g/kg ×5 d). Then Prednisone 1 mg/kg/day (or Dexamethasone 40 mg/day ×4 d), tapered over 4–6 weeks. Avoid NSAIDs and IM injections.	1.5 + 1.5
4	Only for life-threatening bleeding (e.g. intracranial), given with IVIG/steroids, because transfused platelets are destroyed quickly.	1
5	Eltrombopag 50 mg daily or Romiplostim weekly SC (TPO agonists) · Rituximab · Splenectomy (cures ~ ² / ₃ ; vaccinate first).	0.5 each (max 2)

Case 16

Hematology · 10 marks

A 14-year-old boy with sickle cell disease presents with severe pain in his back and legs after a febrile illness and poor fluid intake. Temperature 38.3 °C, mild jaundice, SpO₂ 97%. Hb 7.5 g/dL (baseline 8), reticulocytes 12%.

1. What is the type of crisis and what precipitated it? **[1]**
2. Explain the molecular defect and name 2 smear findings. **[2]**
3. Write the acute management. **[2]**
4. The next day he has chest pain, cough, SpO₂ 88% and a new infiltrate. What has happened, and what is the key treatment? **[2]**
5. **Special:** List his long-term preventive measures, including how hydroxyurea works. Which crisis causes a sudden fall in Hb with **low** reticulocytes? **[3]**

Case 16 — Answers & marking scheme

Q	Model answer	Marks
1	Vaso-occlusive (painful) crisis , precipitated by infection/fever and dehydration.	1
2	β6 Glu→Val : HbS polymerises with hypoxia, acidosis and dehydration → sickling. Smear: sickle cells, Howell-Jolly bodies (target cells). HbS >80% on electrophoresis.	1 + 1
3	IV hydration, O₂ (if hypoxic), adequate analgesia (opioids), treat infection (cultures + antibiotics), keep warm.	2
4	Acute chest syndrome . O ₂ , antibiotics, analgesia + exchange transfusion (target HbS <50%). Exchange is also used for stroke and priapism.	1 + 1
5	Hydroxyurea (↑HbF, target ≥20%; watch neutropenia) · Penicillin V prophylaxis · pneumococcal, Hib and meningococcal vaccines · Folic acid 5 mg daily. Aplastic crisis (parvovirus B19): sudden ↓Hb with reticulocytopenia.	2 + 1

Case 17

Hematology · 10 marks

An 8-year-old boy has a painful, hot, swollen right knee the day after a minor fall. He has had several similar episodes and a large thigh hematoma after a vaccination. His maternal uncle had a "bleeding problem". Platelets $260 \times 10^9/L$, PT normal, aPTT 72 s (control 32 s), bleeding time normal.

1. What is the most likely diagnosis and its inheritance? **[2]**
2. Explain the bleeding pattern (primary vs secondary hemostasis). **[1]**
3. What does a 1:1 mixing study show here, and what would non-correction mean? Which test confirms the diagnosis? **[3]**
4. How do you treat this bleed? **[2]**
5. **Special:** How does von Willebrand disease differ in the lab? When can DDAVP be used? **[2]**

Case 17 — Answers & marking scheme

Q	Model answer	Marks
1	Hemophilia A (factor VIII deficiency) (hemophilia B = factor IX); X-linked recessive .	1 + 1
2	Secondary (coagulation) defect: hemarthroses, deep muscle hematomas, delayed bleeding (primary/platelet defects give petechiae, mucosal and immediate bleeding).	1
3	Prolonged aPTT that corrects on mixing = factor deficiency . No correction = inhibitor : immediate = lupus anticoagulant; after 2 h at 37 °C = factor VIII inhibitor (acquired hemophilia). Confirm with a factor VIII assay .	1 + 1 + 1
4	Recombinant factor VIII concentrate (on demand, then prophylaxis); tranexamic acid / EACA as adjunct; cryoprecipitate or FFP only if concentrate unavailable. Rest, ice, elevation; no IM injections or NSAIDs .	1.5 + 0.5
5	vWD : autosomal dominant; ↑ bleeding time , variable ↑ aPTT , ↓ vWF antigen and ristocetin cofactor ; mucosal bleeding. DDAVP 0.3 µg/kg : mild hemophilia A and vWD (not hemophilia B).	1 + 1

Case 18

Hematology · 10 marks

Ten minutes into a unit of packed red cells, a 50-year-old woman develops fever, rigors, loin pain, chest tightness and a feeling of doom. BP 85/50 mmHg; her urine turns dark red. The nurse finds the unit was meant for another patient in the next bed.

1. What is the reaction and its mechanism? **[2]**
2. What are your immediate steps? **[3]**
3. Name 3 lab findings that confirm intravascular hemolysis. **[1.5]**
4. **Special:** Distinguish TRALI from TACO (2 features each). **[2]**
5. Which reaction is prevented by leukodepletion, and which by gamma-irradiation? **[1.5]**

Case 18 — Answers & marking scheme

Q	Model answer	Marks
1	Acute hemolytic transfusion reaction (<30 min): ABO incompatibility → IgM + complement → intravascular hemolysis.	1 + 1
2	Stop the transfusion , keep the line open with normal saline ; ABC, fluids to maintain urine output; return the bag and giving set to the blood bank ; repeat cross-match, check patient identity; monitor for DIC and AKI.	1 each
3	Any 3: hemoglobinemia, hemoglobinuria , absent haptoglobin, ↑LDH, ↑indirect bilirubin, positive DAT .	0.5 each
4	TRALI (≤6 h): donor anti-HLA/anti-neutrophil antibodies; non-cardiogenic edema, normal PCWP/JVP , often hypotension; ventilate, avoid diuretics . TACO : volume overload; hypertension, ↑JVP, S3, ↑BNP ; responds to furosemide .	1 + 1
5	Leukodepletion → febrile non-hemolytic reaction (FNHTR) . Gamma-irradiation → TA-GVHD (>90% fatal). (Washed cells → anaphylaxis in IgA deficiency.)	0.75 + 0.75

Case 19

Tropical / ID · 10 marks

A 19-year-old university-dorm student has 12 hours of fever, a severe headache and vomiting. Neck stiffness; a petechial rash that does not blanch with the glass test; GCS 14; no focal signs, no papilledema. BP 100/60 mmHg.

1. What is the most likely diagnosis and organism? **[1]**
2. What should happen in the first 30 minutes? When is a CT needed before LP? **[3]**
3. Describe the expected CSF findings. **[2]**
4. Give the empirical antibiotics and the adjunct, with doses and timing. **[2]**
5. **Special:** His roommates are worried. What do you give them? **[2]**

Case 19 — Answers & marking scheme

Q	Model answer	Marks
1	Acute bacterial meningitis with meningococemia: <i>Neisseria meningitidis</i> .	1
2	Do NOT delay antibiotics: take blood cultures → dexamethasone + empirical IV antibiotics immediately , then LP. CT first only if: focal signs, papilledema, GCS ≤10, immunocompromised, seizures . (Kernig/Brudzinski are specific but insensitive, so a negative sign does not exclude meningitis.)	1.5 + 1.5
3	Bacterial: opening pressure >250 mm H₂O ; WBC 1000-10 000/μL , PMN >80% ; glucose <40 mg/dL (CSF:blood <0.4) ; protein >100-500 mg/dL ; lactate >4 ; Gram stain: Gram-negative diplococci.	0.5 each (max 2)
4	Age 18-50: Ceftriaxone 2 g IV q12h + Vancomycin 15-20 mg/kg q8-12h (add Ampicillin 2 g q4h if >50 y / immunocompromised / neonate, for Listeria). Dexamethasone 10 mg IV q6h ×4 days , given 10-20 min before or with the first antibiotic dose.	1 + 1
5	Close-contact chemoprophylaxis (any one): Ciprofloxacin 500 mg once · Rifampin 600 mg bid ×2 days · Ceftriaxone 250 mg IM once . Also notify public health; meningococcal vaccine for dorm residents.	1.5 + 0.5

Case 20

Tropical / ID · 10 marks

A 40-year-old sheep farmer who eats home-made unpasteurised cheese has had 4 weeks of fever that comes and goes in waves, drenching sweats with a mouldy smell, low-back pain and a painful swollen right testis. There is mild hepatosplenomegaly.

1. What is the diagnosis? Name the complication he has. **[2]**
2. List the confirmatory investigations, including any special instruction to the lab. **[2]**
3. Write the preferred regimen with doses and duration, plus an all-oral alternative. **[3]**
4. MRI shows lumbar spondylitis. How does this change treatment? **[1]**
5. **Special:** His pregnant wife is also infected. What do you give her and why? What is the main cause of death in brucellosis? **[2]**

Case 20 — Answers & marking scheme

Q	Model answer	Marks
1	Brucellosis (<i>B. melitensis</i> , from sheep/goat dairy) complicated by epididymo-orchitis (back pain suggests sacroiliitis/spondylitis).	1 + 1
2	Blood / bone-marrow culture: warn the lab (prolonged incubation; lab-acquired infection risk). Serum agglutination test $\geq 1:160$. CBC, LFT.	1 + 1
3	Preferred: Doxycycline 100 mg bid $\times 6$ weeks + Streptomycin 1 g IM daily $\times 14-21$ days (relapse $< 5\%$). Oral alternative: Doxycycline 100 mg bid + Rifampin 600-900 mg daily $\times 6$ weeks (relapse $\sim 10\%$).	2 + 1
4	Complicated disease: Doxycycline + Rifampin + Ceftriaxone 2 g IV q12h (or streptomycin) for 12 weeks to 6 months .	1
5	Pregnancy: TMP-SMX + Rifampin $\times 6$ weeks , avoiding doxycycline (teeth/bone) and aminoglycosides (fetal ototoxicity). Main cause of death: endocarditis (aortic valve) , which needs valve replacement.	1 + 1

Case 21

Tropical / ID · 10 marks

A 32-year-old man had bloody mucoid diarrhea with tenesmus 2 months ago, treated at a pharmacy. He now has 10 days of fever and right upper quadrant pain referred to the right shoulder, with tender hepatomegaly and intercostal tenderness. Ultrasound: a single 7 cm hypoechoic lesion in the right lobe.

1. What is the most likely diagnosis? How did the organism reach the liver? **[2]**
2. Compare it with a pyogenic abscess and a hydatid cyst (aspirate/imaging). **[2]**
3. Write the drug treatment with doses. Why is a second drug needed? **[3]**
4. When is drainage indicated? **[2]**
5. **Special:** Which bloody-diarrhea organism must **not** be treated with antibiotics or loperamide, and why? **[1]**

Case 21 — Answers & marking scheme

Q	Model answer	Marks
1	Amoebic liver abscess (<i>Entamoeba histolytica</i>) following amoebic dysentery (flask-shaped colonic ulcers); trophozoites reach the liver via the portal vein .	1 + 1
2	Amoebic: " anchovy sauce ", odourless, usually solitary right lobe. Pyogenic: yellow-green foul pus, often multiple , biliary source. Hydatid: unilocular cyst, water-lily sign, daughter cysts .	0.75 + 0.75 + 0.5
3	Tissue amoebicide: Metronidazole 750-800 mg tds ×10 days (or Tinidazole 2 g/day ×3 d), then luminal agent: Diloxanide furoate 500 mg tds ×10 days (or Paromomycin). The luminal agent eradicates intestinal cysts and prevents relapse and spread.	1 + 1 + 1
4	Seldom needed (metronidazole cures most ≤10 cm). Drain if: imminent rupture, left-lobe abscess (risk of pericardial rupture), >10 cm , or no response after 4 days .	0.5 each
5	EHEC O157:H7 : antibiotics and loperamide increase Shiga-toxin release → HUS . Hydrate only.	1

Case 22

Tropical / ID · 10 marks

A 28-year-old man presents with 1 week of anorexia, nausea, dark urine and jaundice. ALT 1450 U/L, bilirubin 6 mg/dL, INR 1.1. He had unprotected sex 3 months ago. Serology: HBsAg positive, anti-HBc IgM positive, anti-HCV negative, anti-HAV IgM negative.

1. Interpret his serology. What are the risks of becoming chronic in adults vs neonates? **[2]**
2. Interpret these profiles: (a) HBsAg + for >6 months, anti-HBc IgG +; (b) anti-HBs + and anti-HBc IgG +; (c) anti-HBs + alone. **[3]**
3. Define acute liver failure. Which finding would make you worried in this patient? **[2]**
4. Name the oral drugs for chronic HBV with doses. **[1]**
5. **Special:** His partner is 20 weeks pregnant. Which hepatitis virus carries a 20–25% mortality in pregnancy? Give the HBV vaccine schedule. **[2]**

Case 22 — Answers & marking scheme

Q	Model answer	Marks
1	Acute hepatitis B (HBsAg + anti-HBc IgM). Chronicity: 5-10% in adults, ~90% in neonates .	1 + 1
2	(a) Chronic HBV infection · (b) recovered, immune by past infection · (c) vaccinated .	1 each
3	Acute liver failure: jaundice + INR >1.5 + encephalopathy in a patient without cirrhosis. Warning signs here: rising INR, confusion/drowsiness, hypoglycemia, shrinking liver.	1 + 1
4	Tenofovir 300 mg daily or Entecavir 0.5 mg daily (Peg-IFN α -2a 180 μ g/week $\times$ 48 wk is an alternative).	1
5	Hepatitis E (fecal-oral). HBV vaccine at 0, 1 and 6 months (screen and vaccinate the partner; HBIG + vaccine for the newborn if she is HBsAg positive).	1 + 1

Case 23

Tropical / ID · 10 marks

A 16-year-old boy from a Nile Delta village who swims in irrigation canals complains of painful urination with blood in the last drops of urine. His 45-year-old uncle from the same village was admitted with hematemesis; he has splenomegaly and a shrunken-looking liver on ultrasound with "pipestem" periportal fibrosis.

1. Name the species causing the boy's illness and the uncle's illness. **[2]**
2. List 3 long-term complications of the urinary form. **[1.5]**
3. Give the treatment with dose. Is it safe in pregnancy? **[2]**
4. How would you manage the uncle's variceal bleeding (main lines)? **[1.5]**
5. **Special:** A tourist develops fever, urticaria and eosinophilia 4 weeks after swimming in the Nile. What is this, and how is it treated? **[3]**

Case 23 — Answers & marking scheme

Q	Model answer	Marks
1	Boy: <i>S. haematobium</i> (urinary). Uncle: <i>S. mansoni</i> (intestinal/hepatic: Symmers pipestem fibrosis → presinusoidal portal hypertension, varices).	1 + 1
2	Any 3: bladder calcification , ureteric obstruction/ hydronephrosis , squamous cell carcinoma of the bladder , recurrent UTI, renal failure.	0.5 each
3	Praziquantel 40 mg/kg on one day (single dose or 2 doses 4–6 h apart). Yes, safe in pregnancy.	1.5 + 0.5
4	Resuscitation (IV access, cautious transfusion), vasoactive drug, antibiotics, endoscopic band ligation ; then praziquantel and secondary prophylaxis. (Liver function is usually preserved in schistosomal fibrosis.)	1.5
5	Katayama fever (acute schistosomiasis), 2–8 weeks after exposure. Prednisone 40 mg/day to control the reaction, then praziquantel after the worms mature (it does not kill immature worms).	1 + 1 + 1

Case 24

Tropical / ID · 10 marks

A 19-year-old student has 1 week of fever, severe sore throat with exudative tonsillitis, fatigue and tender bilateral posterior cervical nodes. The spleen tip is palpable. A GP gave ampicillin and she developed a widespread maculopapular rash. Blood film: lymphocytosis with >10% atypical lymphocytes; ALT mildly raised.

1. What is the diagnosis and the cause of the rash? **[2]**
2. Name 2 serological tests that confirm it. **[2]**
3. How do you differentiate it from CMV and toxoplasma mononucleosis (one clinical + one lab point each)? **[2]**
4. Outline management. When are steroids indicated? **[2]**
5. **Special:** What advice do you give about contact sports, and why? **[2]**

Case 24 — Answers & marking scheme

Q	Model answer	Marks
1	Infectious mononucleosis (EBV). Rash: ampicillin/amoxicillin-induced rash in EBV, which is not true penicillin allergy.	1 + 1
2	Monospot (heterophile antibody) positive; VCA IgM positive (EBNA absent acutely).	1 + 1
3	CMV: mild throat, generalised nodes; Monospot –, anti-CMV IgM/PCR +. Toxoplasma: mild/no throat, unilateral non-tender nodes, minimal atypical lymphocytes; Monospot –, anti-Toxo IgM.	1 + 1
4	Supportive: rest, fluids, paracetamol; avoid ampicillin. Prednisolone 40-60 mg x3 days for airway obstruction , severe thrombocytopenia or hemolysis.	1 + 1
5	Avoid contact sports/heavy lifting for several weeks (until the spleen is back to normal size), because of the risk of splenic rupture (splenomegaly in ~50%).	1 + 1

Case 25

Immunology · 10 marks

Five minutes after an IM penicillin injection, a 30-year-old woman develops generalised flushing and hives, a hoarse voice with stridor, wheeze and abdominal cramps. BP 70/40 mmHg, pulse 135/min, weak.

1. What is the diagnosis? Give 2 features that distinguish it from a vasovagal faint. **[2]**
2. Write the immediate management in order, with doses and route. **[4]**
3. Which blood test supports the diagnosis, and when should it be taken? **[1]**
4. **Special:** She improves after 30 minutes and wants to go home. What do you tell her, and what does she leave with? **[2]**
5. What is the immune mechanism? **[1]**

Case 25 — Answers & marking scheme

Q	Model answer	Marks
1	Anaphylaxis (anaphylactic shock). Vs vasovagal (any 2): tachycardia and a weak pulse (vasovagal: bradycardia) · urticaria/flushing/itch (vasovagal: pallor, no hives) · wheeze/stridor · shock that persists when supine.	1 + 1
2	Epinephrine 1:1000 IM, anterolateral thigh, 0.3-0.5 mg (0.01 mL/kg, max 1 mL); repeat every 5-15 min (up to 2 more doses). Never SC. - Lie flat with legs raised (never sit up) · 100% O₂ 8-10 L/min · large-bore IV · saline 1-2 L · monitor every 2-5 min. - Only after epinephrine: Hydrocortisone 250-500 mg IV, Diphenhydramine 25-50 mg + H2 blocker (famotidine), Salbutamol 2.5-5 mg neb for wheeze. - Refractory: IV epinephrine infusion; stridor/tongue edema → early intubation.	1.5 + 1 + 1 + 0.5
3	Serum tryptase within 1-2 hours of onset.	1
4	Keep her under observation for 6-24 h because of biphasic anaphylaxis (recurrence 3-4 h later, in up to 20%). On discharge: epinephrine auto-injector (0.3 mg) with training, a penicillin-allergy label and allergist referral.	1 + 1
5	Type I (IgE-mediated) hypersensitivity: allergen cross-links IgE on mast cells → degranulation (histamine, tryptase, leukotrienes).	1

A 22-year-old woman has recurrent episodes of non-itchy swelling of the lips, hands and once the larynx, each lasting 2–3 days, often after dental work or around her periods. Some attacks cause severe abdominal colic; she once had an unnecessary laparotomy. Antihistamines, steroids and epinephrine did not help. Her father has similar attacks.

1. What is the most likely diagnosis and its inheritance? Which mediator causes the swelling? **[2]**
2. Give 3 features that separate it from mast-cell (allergic) angioedema. **[1.5]**
3. What is the screening test, and how do you tell type 1 from type 2? **[2.5]**
4. Write the treatment of an acute laryngeal attack, and 2 options for prophylaxis. **[2.5]**
5. **Special:** A 60-year-old hypertensive man develops tongue swelling without urticaria. Which drug is the likely cause, and what do you do? **[1.5]**

Case 26 — Answers & marking scheme

Q	Model answer	Marks
1	Hereditary angioedema: autosomal dominant C1-inhibitor deficiency ; mediator = bradykinin .	1 + 0.5 + 0.5
2	Any 3: no urticaria or itch (tension, burning) · severe abdominal colic mimicking a surgical abdomen · triggers are trauma/dental work/stress/menses · no response to epinephrine, steroids or antihistamines .	0.5 each
3	Serum C4 : low even between attacks (a normal C4 during an attack excludes HAE). Then C1-INH level and function : type 1 (85%) low level + low function; type 2 (15%) normal/high level, low function.	1 + 1.5
4	Secure the airway; C1-INH concentrate IV (Berinert, Cinryze, Ruconest) or icatibant 30 mg SC or Ecallantide 30 mg SC. Prophylaxis (any 2): danazol/stanozolol, tranexamic acid (children, pregnancy), regular C1-INH.	1.5 + 1
5	ACE inhibitor (ACE = kininase II → bradykinin accumulates). Stop it permanently and protect the airway; do not switch to another ACEI.	1 + 0.5

A 35-year-old woman has had itchy wheals almost every day for 3 months, sometimes with lip swelling. Each wheal disappears within a few hours and leaves no mark. No clear trigger. She also has Hashimoto thyroiditis. Examination between attacks is normal.

1. What is the diagnosis, and what defines "chronic"? Give the 2 main subtypes. **[2]**
2. What investigations are needed? **[1]**
3. Outline the stepwise treatment ladder (EAACI/GA²LEN) with time frames. **[4]**
4. **Special:** Which features suggest urticarial vasculitis instead, and what would you do? **[2]**
5. What is the emergency treatment if she develops airway swelling? **[1]**

Case 27 — Answers & marking scheme

Q	Model answer	Marks
1	Chronic spontaneous urticaria: wheals ± angioedema for ≥6 weeks , each wheal fading within 24 h. Subtypes: CSU (80%) , often autoreactive (anti-FcεRI/IgE, anti-TPO); chronic inducible (20%) : dermatographism, cold, cholinergic, pressure, solar, aquagenic.	1 + 1
2	No routine tests for straightforward urticaria (limited CBC/CRP at most).	1
3	2nd-generation H1 antihistamine, standard dose (e.g. cetirizine 10 mg) - After 2–4 weeks: up-titrate up to 4× dose - After 2–4 weeks: add Omalizumab (anti-IgE, 300 mg SC monthly) - After 6 months: add Cyclosporine (specialist)	1 each
4	Red flags: wheals lasting >24 h, painful lesions, purpura/residual pigmentation, ↑ESR/CRP, low C3/C4 → skin biopsy and refer.	1 + 1
5	Epinephrine 1:1000 0.5 mL IM + hydrocortisone 100–200 mg IV (as for anaphylaxis).	1

Case 28

Immunology · 10 marks

A 3-year-old boy has had 4 pneumonias, recurrent otitis media and 2 episodes of sinusitis since the age of 7 months, each needing IV antibiotics. He is failing to thrive. Tonsils are barely visible and no lymph nodes are palpable. His maternal uncle died of infection in childhood. Serum IgG, IgA and IgM are all very low; CD19+ B cells are absent.

1. What is the most likely diagnosis? Why did infections start after 6 months of age? **[2]**
2. List 4 Jeffrey Modell warning signs present or relevant here. **[2]**
3. What is the main treatment? **[2]**
4. **Special:** Which vaccines are contraindicated in cellular/combined immunodeficiency? Are inactivated vaccines safe? **[2]**
5. Match: (a) hypocalcemia + cardiac defect + absent thymus; (b) eczema + thrombocytopenia + infections; (c) recurrent *Neisseria* infections; (d) anaphylaxis to blood products. **[2]**

Case 28 — Answers & marking scheme

Q	Model answer	Marks
1	X-linked agammaglobulinemia (Bruton): B-cell defect. Infections start after ~6 months when maternal IgG has waned.	1 + 1
2	Any 4: ≥8 otitis/yr · ≥2 sinusitis · ≥2 pneumonias · need for IV antibiotics · failure to thrive · family history · deep abscesses · ≥2 months of oral antibiotics with little effect.	0.5 each
3	Lifelong IVIG or SCIG replacement , with prompt antibiotics (± prophylactic co-trimoxazole).	2
4	Live vaccines: BCG, OPV, MMR, varicella, rotavirus, yellow fever, intranasal flu. Inactivated vaccines are safe.	1 + 1
5	(a) DiGeorge (22q11) · (b) Wiskott-Aldrich · (c) C5-C9 (terminal complement) deficiency · (d) selective IgA deficiency .	0.5 each

Pearl: HSCT is curative for SCID. Refer any infant with failure to thrive + recurrent infections, absent thymic shadow or lymphocytes <1500/μL.

Case 29

Immunology / Dermatology · 10 marks

A 45-year-old man started allopurinol 2 weeks ago. He now has fever, painful dusky-red macules that are blistering and peeling (about 15% of body surface), positive Nikolsky sign, hemorrhagic crusted lips, red eyes and genital erosions.

1. What is the diagnosis? How is it classified by body-surface area? **[2]**
2. Which hypersensitivity type is this? **[1]**
3. Outline management. **[4]**
4. After recovery, can a drug provocation test confirm the culprit? **[1]**
5. **Special:** How would DRESS differ clinically from this patient? **[2]**

Case 29 — Answers & marking scheme

Q	Model answer	Marks
1	SJS/TEN overlap (15% detached), drug-induced (allopurinol). Detached BSA: SJS <10% • overlap 10-30% • TEN >30% ; ≥2 mucosal sites involved.	1 + 1
2	Type IVc : CD8 cytotoxic T cells (perforin/granzyme, Fas-L), onset 4–28 days.	1
3	• Stop the drug immediately (and any non-essential drugs) • Admit to ICU/burn unit; barrier nursing, wound care • Fluids and electrolytes, nutrition, temperature control • Eye care (ophthalmology), mouth and genital care, lung care • IVIG or cyclosporine; SCORTEN to predict mortality	1 for stopping drug; 0.75 each other (max 4)
4	No : provocation is absolutely contraindicated after SJS/TEN (also DRESS, exfoliative dermatitis). The drug is avoided for life.	1
5	DRESS: later onset (2–6 weeks), fever + widespread rash + internal organ involvement (hepatitis, nephritis, pneumonitis, myocarditis), eosinophils >10% , atypical lymphocytes, lymphadenopathy; no extensive epidermal detachment. Type IVb.	2

Case 30

Venereology · 10 marks

A 25-year-old man has a single painless ulcer on the coronal sulcus that appeared 3 weeks after unprotected sex. The ulcer is firm and indurated with a clean base. He has bilateral rubbery, non-tender, non-suppurating inguinal nodes.

1. What is the most likely diagnosis? Give 2 differential diagnoses and how each differs. **[3]**
2. List the investigations and the role of each. **[3]**
3. Write the treatment, including the option if he is allergic to penicillin. **[2]**
4. **Special:** 6 hours after the injection he has fever, chills and tachycardia. What is happening, and what do you do? **[1]**
5. How do you confirm cure? **[1]**

Case 30 — Answers & marking scheme

Q	Model answer	Marks
1	Primary syphilis (hard chancre). Chancroid: 3–7 d incubation, multiple, very painful, soft ulcers with ragged undermined edges and purulent base; tender suppurative bubo. Genital herpes: multiple grouped painful vesicles → shallow erosions; tender bilateral nodes.	1 + 1 + 1
2	• Darkfield microscopy of chancre exudate: motile spirochetes (immediate diagnosis) • VDRL/RPR (non-treponemal): screening and monitoring • TPHA / FTA-ABS (treponemal): confirmation; stays positive for life (FTA-ABS IgM earliest) • Screen for other STIs including HIV	0.75 each
3	Benzathine penicillin G 2.4 MU IM single dose. Allergy: Doxycycline 100 mg bid ×14 days (Tetracycline 500 mg qid ×15 d; in pregnancy Erythromycin 500 mg qid ×15 d). Treat partners.	1 + 1
4	Jarisch-Herxheimer reaction (endotoxin release from dying spirochetes, 2–24 h), not an allergy. Continue penicillin; give antipyretics ± steroids.	1
5	4-fold fall in VDRL/RPR titre on serial testing (treponemal tests cannot be used to monitor).	1

Pearl: Late latent syphilis (>1 y) needs Benzathine penicillin 2.4 MU IM **weekly ×3** (allergy: Doxycycline ×30 d).

Case 31

Dermatology · 10 marks

A father, his pregnant wife (24 weeks) and their 6-week-old baby all have severe itching that is worse at night. The father has excoriated papules and S-shaped burrows in the finger webs and on the wrists, around the umbilicus and on the genitalia. The baby has lesions on the scalp, palms and soles.

1. What is the diagnosis? Give 2 features that support it. **[2]**
2. Write the father's treatment with application instructions. **[3]**
3. **Special:** Which treatment is safe for the pregnant mother and for the 6-week-old baby? Which drug is contraindicated in both? **[3]**
4. What general measures stop re-infestation? **[2]**

Case 31 — Answers & marking scheme

Q	Model answer	Marks
1	Scabies (<i>Sarcoptes scabiei</i>). Any 2: nocturnal itch · several family members affected · burrows (S-shaped, ending in a vesicle) · typical sites (finger webs, wrists, axillae, umbilicus, genitalia). The head is spared in adults but involved in infants.	1 + 1
2	Permethrin 5% cream from neck to toes , left on for 8-12 h (overnight) then washed off; repeat on day 8 . Alternatives: Benzyl benzoate 25%, Ivermectin 200 µg/kg orally repeated on day 8. Antihistamine for itch, which can last 2-4 weeks after cure.	1 + 1 + 1
3	Precipitated sulfur (10% adult / 5% child) nightly for 3-5 days is safe in pregnancy and infants <2 months ; Permethrin 2.5% for older infants (include scalp and face in infants). Ivermectin is contraindicated in pregnancy and in children <15 kg.	1 + 1 + 1
4	Treat all close contacts at the same time , even without symptoms. Wash clothes and bed linen at >60 °C , or seal them in a bag for 10 days.	1 + 1

Pearl: Crusted (Norwegian) scabies occurs in immunocompromised patients and is highly contagious.

Case 32

Venereology · 10 marks

A 26-year-old man has had burning on urination and a profuse, thick, creamy yellow urethral discharge for 1 day. He had unprotected intercourse with a new partner 3 days ago. Gram stain of the discharge: Gram-negative intracellular diplococci inside neutrophils.

1. What is the diagnosis and the organism? **[1]**
2. How does non-gonococcal urethritis differ (incubation, discharge, organisms, lab)? **[2]**
3. Write the treatment with doses. Why is a second drug added? **[3]**
4. List 2 complications in men and 2 in women. **[2]**
5. **Special:** What is Reiter syndrome? Name one other essential step in management. **[2]**

Case 32 — Answers & marking scheme

Q	Model answer	Marks
1	Gonococcal urethritis: <i>Neisseria gonorrhoeae</i> (Gram-negative kidney-shaped diplococcus, oxidase +).	1
2	NGU: incubation 2-3 weeks ; scanty mucoid "morning drop" ; Chlamydia trachomatis D-K , <i>Mycoplasma genitalium</i> , <i>Ureaplasma</i> ; ≥5 PMN/HPF without diplococci , NAAT.	0.5 each
3	Ceftriaxone 250 mg IM once + Azithromycin 1 g PO once (alternative: Spectinomycin 2 g IM). Azithromycin covers co-existing Chlamydia . Cure = discharge resolves by day 3.	1 + 1 + 1
4	Men (any 2): epididymitis, prostatitis, periurethral abscess, Tysonitis/Cowperitis, urethral stricture . Women (any 2): Bartholinitis, cervicitis, PID → tubal infertility, ectopic pregnancy .	1 + 1
5	Reactive arthritis after Chlamydia: urethritis + conjunctivitis + arthritis ("can't see, can't pee, can't climb a tree"). Treat partner(s) at the same time (and screen for syphilis/HIV).	1 + 1

Pearl: Trichomoniasis: frothy green-yellow discharge, strawberry cervix → Metronidazole or Tinidazole 2 g once; treat partner.

Case 33

Dermatology · 10 marks

A 7-year-old boy (25 kg) has a round, scaly, grey patch of hair loss on the scalp with broken-off hairs. His classmate has a similar lesion. Under Wood's lamp the patch fluoresces bright green.

1. What is the diagnosis and the likely genus? **[2]**
2. Write the treatment with dose and duration. Why are topical antifungals alone not enough? **[3]**
3. What does it mean if there is **no** fluorescence and black dots on the scalp? **[1]**
4. **Special:** The lesion becomes a boggy, pustular, tender swelling. What is it, how do you treat it, and what must you never do? **[3]**
5. Name the Wood's lamp colour of pityriasis versicolor and of erythrasma. **[1]**

Case 33 — Answers & marking scheme

Q	Model answer	Marks
1	Tinea capitis, scaly (grey patch) type: <i>Microsporum</i>.	1 + 1
2	Griseofulvin 12.5 mg/kg/day ×6-8 weeks (≈ 300 mg/day for 25 kg), taken with fatty food. The fungus is inside the hair shaft/follicle, which topical agents do not reach (topicals/shampoo only reduce spread).	2 + 1
3	Black dot ringworm (<i>T. violaceum</i> , endothrix): Griseofulvin 6–8 wk or Terbinafine 250 mg 4–6 wk.	1
4	Kerion (inflammatory reaction). Griseofulvin + prednisolone 1 mg/kg for 10–14 days. Never incise and drain.	1 + 1 + 1
5	Pityriasis versicolor: golden-yellow . Erythrasma: coral red .	0.5 + 0.5

Pearl: Favus = yellow cup-shaped scutula with mousy odour → griseofulvin 8–10 weeks; it leaves scarring alopecia.

Case 34

Dermatology · 10 marks

A 5-year-old girl has had spreading red erosions with golden "honey-coloured", stuck-on crusts around her nose and mouth for 4 days. She is afebrile. Her younger brother has started to develop similar lesions.

1. What is the diagnosis and the organisms? **[2]**
2. Write the treatment of localised disease and of widespread disease (doses). **[3]**
3. **Special:** 3 weeks later she has puffy eyes and cola-coloured urine. What is it? **[1]**
4. Contrast erysipelas with cellulitis (organism, depth, edge) and give a first-line treatment for erysipelas. **[3]**
5. Why should you never squeeze a furuncle on the upper lip or nose? **[1]**

Case 34 — Answers & marking scheme

Q	Model answer	Marks
1	Non-bullous impetigo (contagious): group A streptococcus ± <i>S. aureus</i> . (Bullous impetigo = <i>S. aureus</i> phage II exfoliative toxin, neonates.)	1 + 1
2	Localised: remove crusts (olive oil, warm saline), KMnO ₄ 1/8000, Fusidic acid 2% or Mupirocin 2% tid ×7 days . Widespread: Amox-clav (adult 1 g q12h; weight-based in children) or Flucloxacillin ×7–10 d; allergy: Erythromycin or Azithromycin. School exclusion until crusts dry.	1.5 + 1.5
3	Post-streptococcal glomerulonephritis (2–3 weeks after impetigo).	1
4	Erysipelas : GAS; superficial dermis and lymphatics; fiery red, sharply demarcated raised border ; fever, rigors. Cellulitis : strep or <i>S. aureus</i> ; deep dermis and fat; ill-defined, not raised . Erysipelas: Penicillin V 500 mg q6h ×10–14 d (or procaine penicillin 1.2 MU IM daily; IV penicillin G if severe).	1 + 1 + 1
5	The "danger area of the face" drains to the cavernous sinus , so squeezing risks cavernous sinus thrombosis .	1

Case 35

Dermatology · 10 marks

A 19-year-old woman has had acne for 4 years. She now has painful nodules and cysts on the face and back with early scarring. Topical retinoid, benzoyl peroxide and 3 months of oral doxycycline did not help. She is sexually active.

1. List the 4 pathogenic factors in acne. **[2]**
2. Give the treatment of mild and of moderate acne. Why is topical clindamycin always combined with benzoyl peroxide? **[3]**
3. What is the treatment for her severity (dose, duration, cumulative dose)? **[2]**
4. **Special:** What must be arranged before and during this treatment? Name 3 side effects to monitor. **[3]**

Case 35 — Answers & marking scheme

Q	Model answer	Marks
1	Follicular hyperkeratosis · androgen-driven sebum · <i>Cutibacterium acnes</i> · inflammation.	0.5 each
2	Mild (comedones, few papules): topical retinoid nightly (tretinoin 0.025–0.05%, adapalene 0.1%) ± benzoyl peroxide 2.5–5% ± clindamycin 1%. Moderate: topical retinoid + oral Doxycycline 100 mg daily (or Minocycline 50–100 mg) for 2–4 months. BPO prevents bacterial resistance (no resistance to BPO).	1 + 1 + 1
3	Isotretinoin 0.5–1 mg/kg/day for ~6 months (cumulative 120–150 mg/kg). (Alternative in females: cyproterone acetate 2 mg + ethinylestradiol 35 µg.)	2
4	Teratogenic: pregnancy test and double contraception from 1 month before until 1 month after treatment. Baseline and periodic lipids (TG, cholesterol) and LFT. Side effects (any 3): cheilitis, xerosis, epistaxis, ↑TG/cholesterol, ↑ALT.	1 + 1 + 1

Case 36

Dermatology · 10 marks

A 68-year-old man had 3 days of burning pain over the left side of his chest, then a band of grouped vesicles on a red base appeared along the same area, stopping at the midline. The rash appeared 36 hours ago.

1. What is the diagnosis and the cause? **[1]**
2. Write the antiviral treatment with doses. What is the time window? **[2]**
3. **Special:** What is Hutchinson sign and why does it matter? What is Ramsay Hunt syndrome? **[3]**
4. When should you suspect an underlying immunosuppression? **[1]**
5. Pain persists 2 months after the rash healed. What is it, and how is it treated (doses)? **[3]**

Case 36 — Answers & marking scheme

Q	Model answer	Marks
1	Herpes zoster (shingles): reactivation of latent VZV in the dorsal root ganglion.	1
2	Within 72 h of rash onset: Acyclovir 800 mg 5×/day ×7–10 d , or Valacyclovir 1 g tds, or Famciclovir 500 mg tds ×7 d. Plus analgesia.	1.5 + 0.5
3	Hutchinson sign: vesicles on the tip of the nose (nasociliary branch of V1) → high risk of herpes zoster ophthalmicus / keratitis → urgent eye review. Ramsay Hunt: facial palsy + ear vesicles + loss of taste (geniculate ganglion).	1.5 + 1.5
4	Bilateral or multidermatomal / disseminated zoster , or young patient → look for HIV, lymphoma, immunosuppression.	1
5	Post-herpetic neuralgia (pain >1 month). Amitriptyline 10–25 mg at night , Gabapentin 300 mg tds , or Pregabalin.	1 + 2

A 30-year-old man and his wife have not conceived after 2 years of regular unprotected intercourse; her work-up is normal. He has gynecomastia, sparse body hair, long limbs and small, firm testes (3 mL each by orchidometer). Semen analysis (×2): volume 2 mL, no sperm even in the centrifuged pellet.

1. Define infertility. Name the semen finding. **[1.5]**
2. Give the WHO 2010 lower reference limits for concentration, total motility, progressive motility and morphology. **[2]**
3. What is the likely diagnosis? Which hormone and test confirm it? **[2]**
4. **Special:** Classify azoospermia as obstructive vs non-obstructive vs hypogonadotropic (FSH, testis size, example, treatment). **[3]**
5. Another man has a "bag of worms" on the left side of the scrotum and OAT. What is it, why is it left-sided, and what is the treatment? **[1.5]**

Case 37 — Answers & marking scheme

Q	Model answer				Marks
1	No pregnancy after 12 months of regular unprotected intercourse (male factor $\approx 50\%$). Finding: azoospermia .				1 + 0.5
2	Concentration ≥ 15 M/mL · total motility $\geq 40\%$ · progressive motility $\geq 32\%$ · morphology $\geq 4\%$ (also volume ≥ 1.5 mL, total ≥ 39 M, vitality $\geq 58\%$).				0.5 each
3	Klinefelter syndrome (47,XXY) : non-obstructive azoospermia. $\uparrow$ FSH (and LH, low testosterone); confirm with karyotype .				1 + 1
4		FSH	Testes	Example $\rightarrow$ treatment	1 per row
	Obstructive	Normal	Normal size	CBAVD (CFTR) $\rightarrow$ vaso-/epididymovasostomy, sperm retrieval	
	Non-obstructive	$\uparrow$	Small	Klinefelter $\rightarrow$ micro-TESE + ICSI	
	Hypogonadotropic	$\downarrow$ (LH $\downarrow$)	Small, soft	Kallmann (anosmia) $\rightarrow$ hCG + hMG	
5	Varicocele (grade III, visible). Left: the left testicular vein drains at a right angle into the left renal vein . Clinical varicocele + abnormal semen $\rightarrow$ microsurgical varicocelectomy .				0.5 each

Case 38

Psychiatry · 10 marks

A 62-year-old man, widowed 6 months ago, has had 2 months of low mood, loss of interest, early-morning waking, 7 kg weight loss, poor concentration and guilt. He says his children "would be better off without me" and admits he has been saving his sleeping tablets. He has no history of mania. He had an MI last year and takes a β -blocker.

1. What is the diagnosis? Justify it using DSM-5 criteria. **[2]**
2. Which medical and drug causes should be excluded, and with which tests? **[2]**
3. **Special:** List his suicide risk factors and the immediate safety steps. **[3]**
4. Give the drug treatment with dose, and state when ECT is indicated. **[3]**

Case 38 — Answers & marking scheme

Q	Model answer	Marks
1	Major depressive disorder (severe) with suicidal ideation. ≥5 of 9 symptoms for ≥2 weeks, including depressed mood or anhedonia: mood, anhedonia, sleep, weight, concentration, guilt, suicidal thoughts. Duration and severity exceed normal grief.	1 + 1
2	Exclude: hypothyroidism (TSH) , diabetes (glucose), post-MI/stroke depression, anemia (CBC), B12; drugs: antihypertensives (β-blocker) , steroids , alcohol/substances. Ask about past mania or hypomania (bipolar: avoid antidepressant monotherapy).	1 + 1
3	Risk factors (any 4): male, elderly, widowed, severe depression, hopelessness, a plan with stockpiled pills , chronic illness. Actions: take every threat seriously ; calm, private interview + collateral history; remove means (pills, sharps, belts); 1:1 observation / admission ; document every step.	1.5 + 1.5
4	SSRI: Sertraline 50-200 mg/day (safest after MI), or Fluoxetine 20 mg / Citalopram 20-60 mg; + psychotherapy (CBT). Response lag 2-4 weeks; review often. ECT (6-12 sessions, 3/week): severe suicidality, stupor, food refusal, psychotic features, treatment resistance.	1.5 + 1.5

A 22-year-old man has withdrawn from university over the past 8 months. He hears two voices commenting on his actions and believes his neighbours are controlling his thoughts through the TV. His speech is hard to follow and his affect is blunted. He neglects hygiene. He denies drug use; urine drug screen is negative.

1. What is the diagnosis? Justify it with DSM-5 criteria (symptoms + duration). **[2]**
2. Name his positive and negative symptoms. **[1]**
3. List 3 differentials to exclude and the tests used. **[2]**
4. Give first-line drug treatment with dose and the baseline tests before starting. **[2]**
5. **Special:** On day 5 of haloperidol he is febrile (40 °C), rigid, sweaty, confused, with labile BP and CPK 8000. What is it, and how is it managed? **[3]**

Case 39 — Answers & marking scheme

Q	Model answer	Marks
1	Schizophrenia. Criterion A: ≥ 2 symptoms for ≥ 1 month (at least one of delusions, hallucinations, disorganised speech) — he has hallucinations, delusions, disorganised speech, negative symptoms; B: marked functional decline; C: continuous disturbance ≥ 6 months.	1 + 1
2	Positive: auditory hallucinations, delusions of control/persecution, disorganised speech. Negative: blunted affect, withdrawal, avolition/self-neglect.	0.5 + 0.5
3	Any 3: substance-induced psychosis (cannabis, amphetamine, tramadol, steroids; urine drug screen) · mood disorder with psychosis (bipolar/psychotic depression; history) · delirium/organic (encephalitis, epilepsy, tumour; CBC, LFT, RFT, MRI, EEG, CSF). Investigations only exclude organic causes; diagnosis is clinical.	2
4	Atypical antipsychotic: Risperidone 4–6 mg/day, Olanzapine 10–20 mg, Quetiapine 300–600 mg or Aripiprazole 15–30 mg. Baseline: ECG, CBC, fasting glucose, lipids (weight).	1 + 1
5	Neuroleptic malignant syndrome. Stop the antipsychotic, admit to ICU, cooling, IV fluids (protect kidneys from rhabdomyolysis), dantrolene or bromocriptine.	1 + 2

Case 40

Psychiatry · 10 marks

A 27-year-old man has slept 2 hours a night for 10 days but feels full of energy. He talks rapidly, jumps between topics, says he will become president, and has spent all his savings on cars. He had a depressive episode 2 years ago. He is irritable when interrupted.

1. What is the diagnosis? Justify it (duration + 3 features). **[2]**
2. How does bipolar II differ? **[1]**
3. Outline acute and maintenance treatment. When is prophylaxis indicated? **[3]**
4. **Special:** Months later on lithium he develops vomiting, diarrhea, coarse tremor, ataxia and slurred speech after starting a diuretic. What is it, and how is it managed? **[3]**
5. Why should antidepressants be avoided alone in his future depressive episodes? **[1]**

Case 40 — Answers & marking scheme

Q	Model answer	Marks
1	Bipolar I disorder, current episode mania: elevated/irritable mood ≥ 1 week (any duration if hospitalised) + ≥ 3 of: $\downarrow$ need for sleep, pressured speech, flight of ideas, grandiosity, reckless spending , distractibility, $\uparrow$ activity.	1 + 1
2	Bipolar II: hypomania (≥ 4 days, no marked impairment/psychosis) + major depression ; never full mania.	1
3	Acute: antipsychotic (olanzapine, risperidone, quetiapine) for rapid control $\pm$ lithium (first-line, 80% response, 1–2 week lag) or valproate/carbamazepine. Maintenance: lithium/mood stabiliser + psychoeducation, social-rhythm therapy. Prophylaxis if ≥ 2 episodes in 5 years , or 1 severe episode + family history.	1 + 1 + 1
4	Lithium toxicity (thiazides $\downarrow$ lithium clearance). Stop lithium , check level, IV saline $\pm$ sodium bicarbonate; hemodialysis is definitive (severe toxicity, encephalopathy, convulsions, renal failure).	1 + 2
5	Antidepressant monotherapy can switch him into mania / rapid cycling; use with a mood stabiliser (or quetiapine).	1

A 26-year-old woman has visited the ED 4 times in 2 months with sudden attacks of palpitations, chest tightness, breathlessness, tingling fingers, dizziness and a feeling she is about to die. Each attack peaks within 10 minutes and settles within 30. ECGs and troponins were normal. She now avoids going out alone in case an attack happens.

1. What is the diagnosis? Give the DSM-5 criteria. What co-existing problem is suggested by her avoidance? **[2]**
2. List 4 medical conditions or substances that can mimic panic attacks, and a test for one. **[2]**
3. How do you manage an acute attack and the long-term condition? **[3]**
4. **Special:** How does generalised anxiety disorder differ? What is the difference between anxiety and phobia? **[2]**
5. Which risk must be assessed in every patient with panic disorder? **[1]**

Case 41 — Answers & marking scheme

Q	Model answer	Marks
1	Panic disorder: recurrent unexpected attacks peaking within minutes + ≥1 month of worry about attacks or behaviour change. Avoidance suggests agoraphobia .	1 + 1
2	Any 4: pheochromocytoma (24-h urine catecholamines/metanephrines) · hyperthyroidism (TSH) · hypoparathyroidism · B12 deficiency · cardiac arrhythmia/ischemia · caffeine, cocaine, amphetamines , bronchodilators · alcohol/BZD withdrawal.	1.5 + 0.5
3	Acute: reassurance , breathing control, a short-acting benzodiazepine only if needed. Long-term: SSRI/SNRI (start low, as SSRI start can worsen anxiety) + CBT with interoceptive exposure ; avoid long-term BZDs.	1 + 2
4	GAD: excessive, uncontrollable, constant worry about many things for ≥6 months + tension, restlessness, fatigue, poor sleep (not discrete attacks). Anxiety = diffuse apprehension with a vague threat; phobia = fear of a specific known object or situation.	1 + 1
5	Suicide risk.	1

A 16-year-old girl is brought by her mother after losing 12 kg in 6 months; BMI is now 15. She exercises for hours, eats very little and says she "still has a fat stomach". Her periods stopped 4 months ago. Pulse 46/min, BP 85/55 mmHg, temperature 35.6 °C, fine downy hair on her arms.

1. What is the diagnosis? State the core criteria and the 2 subtypes. **[2]**
2. List 4 physical or lab complications. **[2]**
3. **Special:** She is admitted for feeding. On day 3 she develops arrhythmia and heart failure. What is this, which electrolyte is key, and how is it prevented? **[3]**
4. Outline long-term management. **[1]**
5. How does bulimia nervosa differ (2 clinical signs + typical acid-base change), and which drug is approved for it? **[2]**

Case 42 — Answers & marking scheme

Q	Model answer	Marks
1	Anorexia nervosa: restriction → low weight (BMI <18.5) , intense fear of weight gain, distorted body image . Subtypes: restricting vs binge-eating/purging .	1 + 1
2	Any 4: bradycardia, hypotension, hypothermia, lanugo, amenorrhea , leukopenia, osteoporosis , electrolyte disturbance.	0.5 each
3	Refeeding syndrome: key electrolyte phosphate (hypophosphatemia) . Prevent by refeeding slowly and monitoring phosphate (and K, Mg), with supplementation.	1 + 1 + 1
4	Nutritional rehabilitation + family-based therapy and CBT.	1
5	Bulimia: binges + compensation ≥1/week for 3 months with normal/high weight ; Russell sign (knuckle calluses), parotid hypertrophy , enamel erosion; hypokalemic hypochloremic metabolic alkalosis . Fluoxetine 60 mg/day (with CBT).	1 + 0.5 + 0.5

Case 43

Psychiatry · 10 marks

On day 3 after a hernia repair, a 48-year-old man becomes acutely confused and agitated, sees insects on the walls, and has a coarse tremor, sweating, temperature 38.2 °C, pulse 124/min and BP 165/100 mmHg. His wife says he drinks a bottle of spirits a day. He had a brief fit yesterday.

1. What is the diagnosis? Explain how the timeline fits. **[2]**
2. What features show this is delirium rather than dementia? Name 3 investigations. **[2]**
3. Write the management with doses. **Special:** what must be given before any glucose, and why? **[4]**
4. Name the triad of Wernicke encephalopathy and 2 drugs for relapse prevention. **[2]**

Case 43 — Answers & marking scheme

Q	Model answer	Marks
1	Alcohol withdrawal → delirium tremens (mortality 10–20%). Timeline from the last drink: 6–12 h tremor/anxiety/sweating → 12–48 h seizures → 72–96 h DTs (clouded consciousness, vivid visual hallucinations, autonomic hyperactivity).	1 + 1
2	Delirium: acute onset over hours, fluctuating, clouded consciousness / impaired attention , worse at night (dementia: insidious, clear consciousness until late). Tests (any 3): glucose, electrolytes incl. Mg and phosphate , LFT, CBC, infection screen, CT brain if head injury or focal signs.	1 + 1
3	• Thiamine 100 mg parenteral bid ×3–5 days, BEFORE any glucose: glucose without thiamine can precipitate Wernicke encephalopathy. Then oral thiamine for ≥6 months. • Benzodiazepine: Chlordiazepoxide 50–100 mg PO q4h or Lorazepam/Diazepam IV (BZDs are the treatment of choice here). • IV fluids, correct electrolytes (K, Mg), quiet lit room, reorientation, no restraints.	2 + 1.5 + 0.5
4	Wernicke: ataxia, confusion, ophthalmoplegia (Korsakoff = amnesia + confabulation). Relapse prevention (any 2): Naltrexone, Acamprosate, Disulfiram .	1 + 1

Pearl: In other kinds of delirium, avoid benzodiazepines (use Haloperidol 2–5 mg IM). Alcohol or sedative withdrawal is the exception.